

Reflection of an Electromagnetic Wave from a Moving Mirror

Source: D. Tong, *Electromagnetism*, Problem Sheet 3, Question 7.

An electromagnetic wave is reflected by a perfect conductor at $x = 0$. The electric field has the form

$$\mathbf{E}(t, \mathbf{x}) = \hat{\mathbf{y}} [f(t_-) - f(t_+)], \quad (1.1)$$

where f is an arbitrary function and $ct_{\pm} = ct \pm x$. Show that this satisfies the relevant boundary condition at the conductor. Find the corresponding magnetic field $\mathbf{B}$.

Show that under a Lorentz transformation to a frame moving with speed v in the x -direction, the electric field is transformed to

$$\mathbf{E}'(t', \mathbf{x}') = \hat{\mathbf{y}} \left[\rho f(\rho t'_-) - \frac{1}{\rho} f\left(\frac{t'_+}{\rho}\right) \right], \quad \text{where } \rho = \sqrt{\frac{c-v}{c+v}}. \quad (1.2)$$

Hence for an incident wave $\mathbf{E}(t, \mathbf{x}) = \hat{\mathbf{y}} F(t_-)$, find the wave that is reflected after it hits a perfectly conducting mirror moving with speed v in the x -direction.

Solution. It is clear to see that the electric field of this form has an ingoing part, $f(t_-)$, and an outgoing part, $f(t_+)$. The boundary condition at a perfect conductor is given by

$$\hat{\mathbf{x}} \cdot (\mathbf{E}_+ - \mathbf{E}_-) = \frac{\sigma}{\epsilon_0} = 0, \quad \text{and} \quad \hat{\mathbf{x}} \times (\mathbf{E}_+ - \mathbf{E}_-) = 0. \quad (1.3)$$

It is clear to see that (1.1) satisfies the boundary condition. To find the magnetic field, let's first take the curl of the electric field, which gives

$$\nabla \times \mathbf{E} = -\hat{\mathbf{y}} \times [\nabla f(t_-) - \nabla f(t_+)] = \frac{\hat{\mathbf{y}} \times \hat{\mathbf{x}}}{c} [\dot{f}(t_-) + \dot{f}(t_+)] = -\frac{\hat{\mathbf{z}}}{c} [\dot{f}(t_-) + \dot{f}(t_+)]. \quad (1.4)$$

Hence, the magnetic field is given by

$$\mathbf{B}(t, \mathbf{x}) = \frac{\hat{\mathbf{z}}}{c} [f(t_-) + f(t_+)]. \quad (1.5)$$

For a boost in the x -direction, the electric field transforms like

$$E'_x = E_x, \quad E'_y = \gamma(E_y - vB_z), \quad \text{and} \quad E'_z = \gamma(E_z + vB_y). \quad (1.6)$$

Therefore, we have

$$\mathbf{E}'(t', \mathbf{x}') = \gamma[f(t_-) - f(t_+)]\hat{\mathbf{y}} - \frac{v}{c}\gamma[f(t_-) + f(t_+)]\hat{\mathbf{y}} = \hat{\mathbf{y}}[\rho f(t_-) - \frac{1}{\rho}f(t_+)]. \quad (1.7)$$

Now, we express the coordinates in the primed coordinates, which is

$$ct_{\pm} = \gamma \left(ct' + \frac{vx'}{c^2} \pm x' \pm vt' \right) = \frac{\gamma}{c} [(c \pm v)t' \pm (c \pm v)x'] = \begin{cases} \rho t'_-, \\ \frac{1}{\rho} t'_+. \end{cases} \quad (1.8)$$

Thus, we find

$$\mathbf{E}'(t', \mathbf{x}') = \hat{\mathbf{y}} \left[\rho f(\rho t'_-) - \frac{1}{\rho} f\left(\frac{t'_+}{\rho}\right) \right]. \quad (1.9)$$

In the frame where the mirror is stationary, the incident electric field is given by $\hat{\mathbf{y}}\rho F(\rho t'_-)$. Since the mirror is stationary in this frame, the reflected electric field must be $-\hat{\mathbf{y}}\rho F(\rho t'_+)$. Given the transformation law we just derived, the reflected electric field in the rest frame is $-\hat{\mathbf{y}}\rho^2 F(\rho^2 t_+)$. The total electric field is given by

$$\mathbf{E}(t, \mathbf{x}) = \hat{\mathbf{y}} [F(t_-) - \rho^2 F(\rho^2 t_+)]. \quad (1.10)$$